

AMMAN, Jordan

WMO: 402700

Lat: 31.973N

Lon: 35.992E

Elev: 2555

StdP: 13.39

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 35.2	(c) 37.3	(d) 12.5	(e) 11.4	(f) 53.5	(g) 17.7	(h) 14.7	(i) 57.9	(j) 29.2	(k) 44.3	(l) 24.1	(m) 45.6	(n) 5.9	(o) 240	(p) 0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.9	(c) 97.0	(d) 64.9	(e) 94.7	(f) 64.8	(g) 91.8	(h) 64.3	(i) 71.7	(j) 86.9	(k) 70.0	(l) 85.1	(m) 68.6	(n) 83.8	(o) 8.2	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 67.3	(b) 110.5	(c) 78.1	(d) 65.4	(e) 103.3	(f) 76.5	(g) 63.8	(h) 97.3	(i) 75.4	(j) 37.2	(k) 85.9	(l) 35.7	(m) 84.8	(n) 34.4	(o) 84.1	(p) 78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.8	(b) 18.6	(c) 16.1	(e) 31.2	(f) 102.9	(g) 2.6	(h) 3.2	(i) 29.3	(j) 105.2	(k) 27.8	(l) 107.1	(m) 26.3	(n) 108.8	(o) 24.4	(p) 111.1		
(4) 21.8	(5) 18.6	(6) 16.1	(7) 31.2	(8) 102.9	(9) 2.6	(10) 3.2	(11) 29.3	(12) 105.2	(13) 27.8	(14) 107.1	(15) 26.3	(16) 108.8	(17) 24.4	(18) 111.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	47.6	49.9	55.4	62.9	71.5	76.8	80.4	80.2	76.5	70.2	59.3	50.8
	DBStd	13.06	4.73	5.69	7.02	7.60	6.72	5.11	4.25	4.00	4.37	5.67	5.91	4.96
	HDD50	242	102	63	23	2	0	0	0	0	0	0	5	47
	HDD65	2067	540	425	309	130	17	0	0	0	0	17	188	441
	CDD50	5792	27	59	191	390	667	804	941	936	796	626	284	71
	CDD65	2141	0	1	12	67	219	354	476	471	346	178	17	0
	CDH74	21630	0	4	117	661	2264	3713	5224	5027	3161	1352	105	2
CDH80	9429	0	0	22	191	927	1650	2480	2376	1324	447	12	0	
Wind	WSAvg	6.1	5.9	6.4	6.7	6.4	6.8	7.5	8.0	6.8	5.4	4.3	4.4	5.0
Precipitation	PrecAvg	10.7	2.5	2.0	1.9	0.6	0.1	0.0	0.0	0.0	0.0	0.3	1.0	2.1
	PrecMax	16.9	9.3	5.9	6.1	5.9	0.9	0.0	0.0	0.0	0.1	1.4	5.2	6.8
	PrecMin	3.8	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.6	1.7	1.4	1.4	1.1	0.2	0.0	0.0	0.0	0.0	0.4	1.1	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.5	73.5	82.7	89.9	97.1	98.7	102.0	100.6	98.3	92.9	82.0	71.6
		MCWB	49.6	52.7	57.1	59.4	62.8	64.1	68.0	66.9	64.9	62.1	56.5	50.3
	2%	DB	62.2	68.1	77.0	84.9	92.9	95.1	96.7	95.3	93.6	88.2	77.0	65.9
		MCWB	48.6	50.4	54.9	57.5	61.6	63.4	66.4	66.3	63.9	61.0	55.4	50.3
	5%	DB	58.9	63.4	72.0	81.2	89.2	91.6	93.2	93.0	90.0	85.0	73.2	62.3
		MCWB	47.6	49.0	52.3	56.3	60.1	62.8	65.2	66.3	63.5	60.5	54.7	49.3
	10%	DB	55.7	59.6	68.1	77.2	85.6	88.2	90.5	90.0	87.5	81.3	69.6	59.0
		MCWB	46.4	47.5	51.0	54.8	59.6	62.7	65.1	66.0	63.5	59.7	53.8	48.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	55.5	59.6	63.9	66.4	70.0	73.9	74.4	71.7	67.4	61.2	56.0
		MCDB	60.9	70.1	76.7	85.6	88.5	87.1	90.8	89.0	83.2	80.3	69.0	63.6
	2%	WB	50.4	52.3	56.5	60.1	64.1	68.0	71.5	72.0	69.4	65.3	59.0	53.1
		MCDB	58.0	63.7	71.4	78.0	85.2	85.8	88.2	86.7	82.3	78.1	67.3	60.3
	5%	WB	49.0	50.3	54.3	58.0	62.4	66.2	69.5	70.5	67.9	63.8	57.5	51.5
		MCDB	55.9	59.8	68.1	75.9	83.4	84.4	85.9	85.3	81.4	76.4	66.8	58.3
	10%	WB	47.7	48.8	52.4	56.1	60.7	64.7	68.0	69.0	66.5	62.4	56.0	50.1
		MCDB	54.0	57.3	65.1	74.0	81.0	83.2	84.7	83.7	80.1	74.6	65.7	56.3
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.2	17.3	20.3	21.4	21.1	19.9	20.5	20.7	19.5	17.0	14.4
		MCDBR	19.3	22.2	23.6	25.1	25.6	24.1	22.3	22.3	23.9	23.6	21.5	19.6
		MCWBR	10.1	10.8	11.1	10.9	10.6	10.7	11.1	11.1	9.9	9.5	9.6	10.2
	5% WB	MCDBR	16.2	18.3	20.6	22.8	22.5	21.6	20.6	21.2	20.8	19.9	17.2	15.3
		MCWBR	9.5	10.0	11.1	11.3	11.0	11.5	11.6	11.6	10.2	9.5	9.2	9.1
Clear-Sky Solar Irradiance	taub	0.344	0.373	0.417	0.461	0.460	0.389	0.396	0.403	0.397	0.416	0.366	0.346	
	taud	2.265	2.168	2.027	1.913	1.943	2.178	2.167	2.150	2.168	2.100	2.225	2.270	
	Ebn at Noon	274	277	272	265	265	284	281	278	275	258	263	266	
	Edh at Noon	35	42	52	61	60	47	48	48	45	45	36	33	
All-Sky Solar Radiation	RadAvg	979	1233	1679	2042	2376	2638	2558	2345	2004	1527	1142	917	
	RadStd	59	94	95	94	99	37	30	33	50	59	58	56	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.86	N/A	-4.84	N/A	-1.76	-2.48	N/A	-153	+283
(47) Regional (17 neighbors)		+0.79	N/A	-3.51	N/A	N/A	N/A	-108	+252	+166

Nomenclature: See separate page